3	**(a)**	The gravitational field strength at a point is the gravitational force exerted per unit mass placed at that point.	B1
	(b)	Gravitational force on the satellite provides the centripetal force for the circular motion of the satellite. $\frac{GMm}{r^2} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2$	C1
		$r^3 = \frac{GM}{4\pi^2}T^2$ $= \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{4\pi^2}(94\times60)^2$	C1
		$r = 6.9 \times 10^6$ m	A1
	(c)(i)	From $\frac{GMm}{r^2} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2$ $r^3\omega^2$ = constant or $\frac{r^3}{T^2}$ = constant $\frac{(6.9 \times 10^6)^3}{(94)^2} = \frac{r^3}{(150)^2}$ $r = 9.42 \times 10^6$ m $v = r\omega$ $= (9.42 \times 10^6)\left(\frac{2\pi}{150 \times 60}\right)$ $= 6600$ m s^{-1} (to 2 s.f.)	M1 M1 M1 A0
	(c)(ii)	Gravitational potential energy $U = -\frac{GMm}{r}$ Change in gravitational potential energy $= U_{final} - U_{initial}$ $= -GMm\left(\frac{1}{r_{final}} - \frac{1}{r_{initial}}\right)$ $= -(6.67 \times 10^{-11})(6.0 \times 10^{24})(1200)\left(\frac{1}{9.4 \times 10^6} - \frac{1}{6.9 \times 10^6}\right)$ $= 1.9 \times 10^{10}$ J	 C1 A1